

The insulin-like growth factor-I receptor is regulated by CD28 and protects activated T cells from apoptosis.

Co-stimulatory signals through the CD28 receptor enhance the survival of T cells that have their antigen receptor (TCR) engaged. Here we show that stimulation through the CD28 receptor in the absence of TCR engagement with either an anti-CD28 cross-linking antibody or the CD80 ligand transiently increases expression of the insulin-like growth factor-I receptor (IGF-IR) on T cells. Antibodies that block signaling through the IGF-IR decrease the survival of T cells activated through the TCR and CD28 in the presence of IL-2 by more than 50%, and also enhance susceptibility to Fas-induced apoptosis. CD28 stimulation increases IGF-IR expression on Jurkat cells, and exogenously added IGF-I can protect these cells from Fas-induced apoptosis. We conclude that CD28-mediated enhancement of IGF-IR expression provides activated T cells with essential survival signals that are independent of survival mediated by IL-2 and Bcl-xl.